

TECHNOLOGY STACK ANALYSIS

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Core Technology Selection

The technology stack for OptiCrop was selected to support fast execution, simple deployment, and data visualization:

Technology Component	Specific Choice	Role in Project
Opticrop Technology Stack	Python 3.10+	Data manipulation and model inference.
Programming Language	Python 3.10+	
Web Framework	Flask 3.0.0	Maintains routes, handles forms, and renders templates.
Language : Python 3.10+		
Framework : Flask 3.1.3 - routing, Jinja2 templates, REST endpoints		StandardScaler, RandomForest, and KMeans model pipeline.
Machine Learning : Scikit-learn 1.9 - Random Forest (classifier + regressor), K-Means		
Data Structures : Pandas 3.0, NumPy 2.5		Data frame cleaning, formatting, and numeric operations.
Data : Pandas 3.0, NumPy 2.5		
Graphics Engine : Matplotlib 3.11, Seaborn 0.13		Generates performance evaluation plots.
Plotting : Matplotlib 3.11, Seaborn 0.13		Accuracy charts, confusion matrices.
Frontend : HTML5, CSS3 (glassmorphism), Bootstrap 5		Styles the dark-themed glassmorphic user dashboard.
Frontend CSS : Bootstrap 5 & Vanilla CSS		
Testing : pytest / Python unittest		
Version Ctrl: Git / GitHub		

2. Design Architecture Rationale

Flask was chosen as the web frame because it integrates easily with Python-based scikit-learn models. A local CSV and SQLite database storage layout was selected to avoid external dependencies, keeping the application lightweight and simple to run.